Rotary Wing Aerodynamics (AE 667)

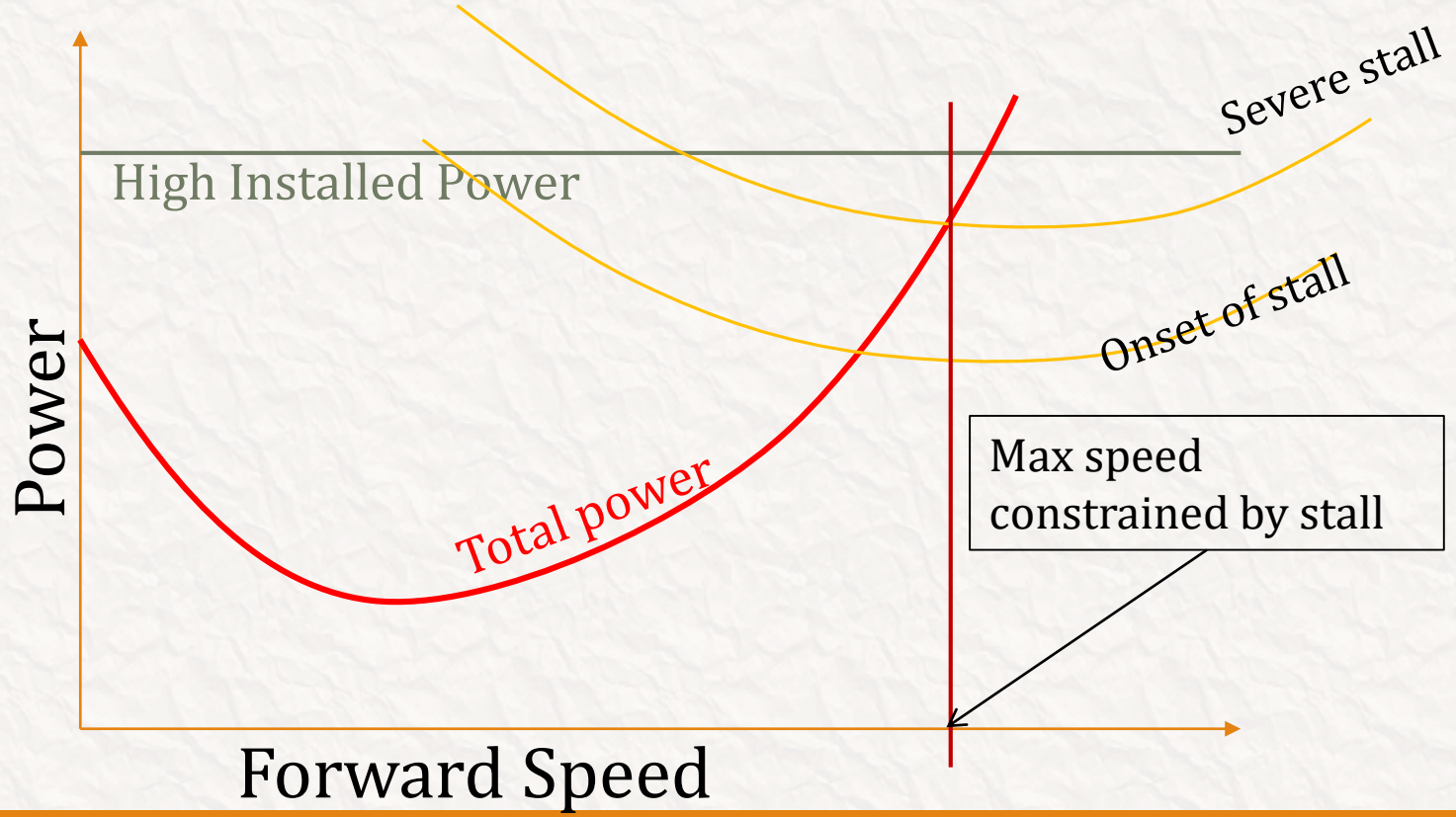


Airfoils for Rotor Blades

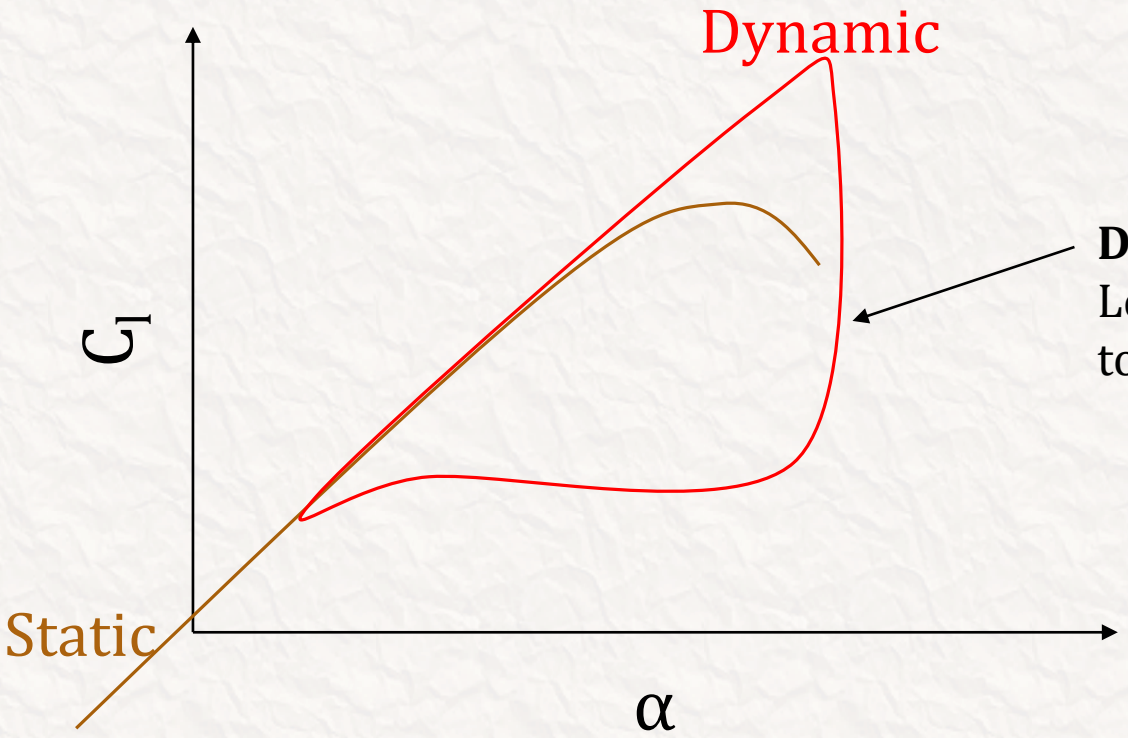
Requirements

- High
  - Static & Dynamic C_l: for high manoeuvre loads
  - Drag divergence number (M_{DD}): to keep high speed flight power requirements under control
- Low
  - Drag Coefficient C_d : to minimize profile power
  - Pitching Moments : for keeping structural forces and vibrations low
- Sufficient thickness for structural strength

Max speed limited by stall ($C_{l,max}$)

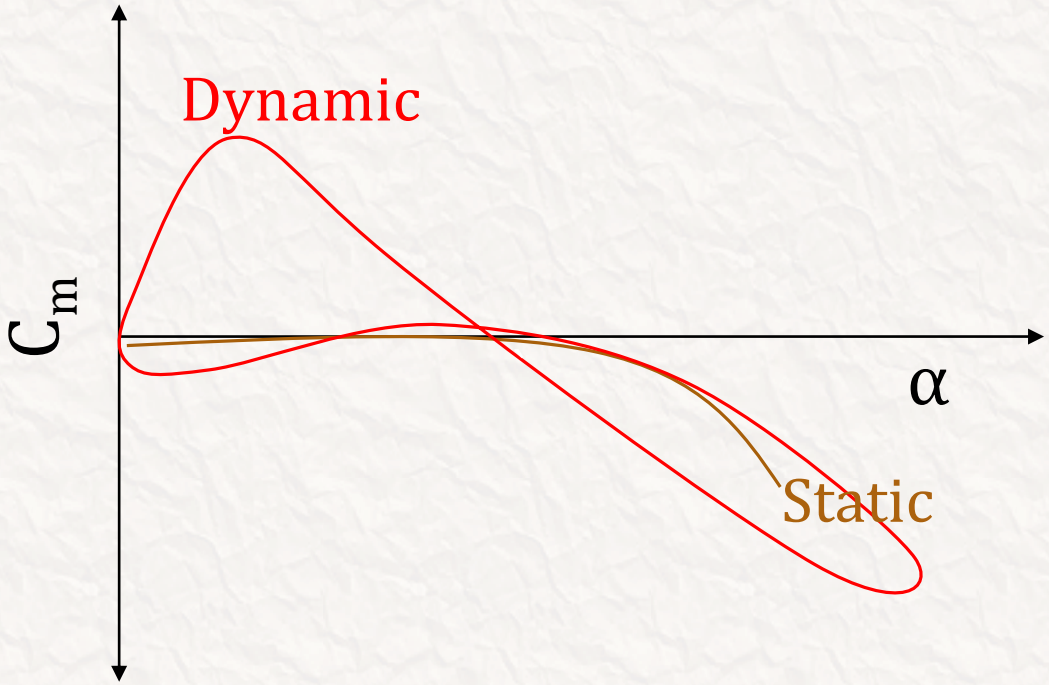


Static and Dynamic lift



Dynamic Stall
Leads to vibration due to large change in lift

Static and Dynamic Pitching Moment

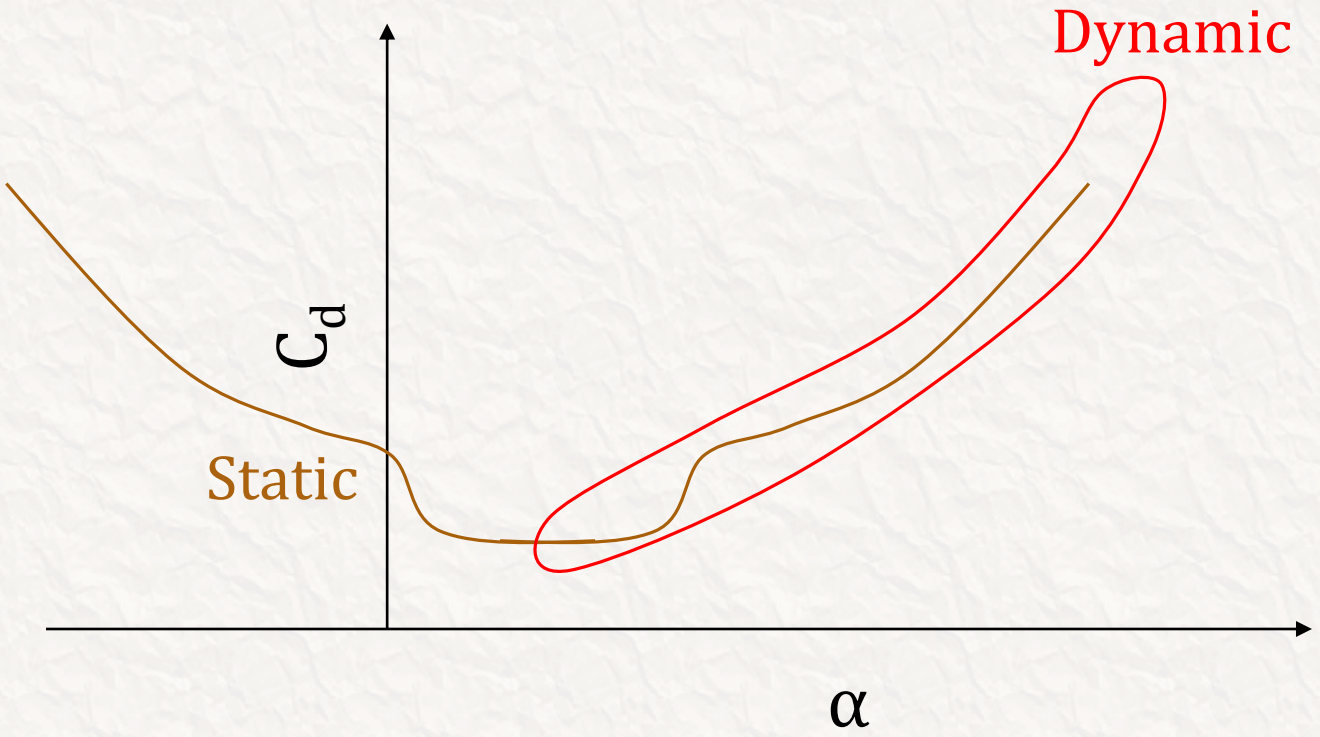


High negative pitching moments fatigue pitch links

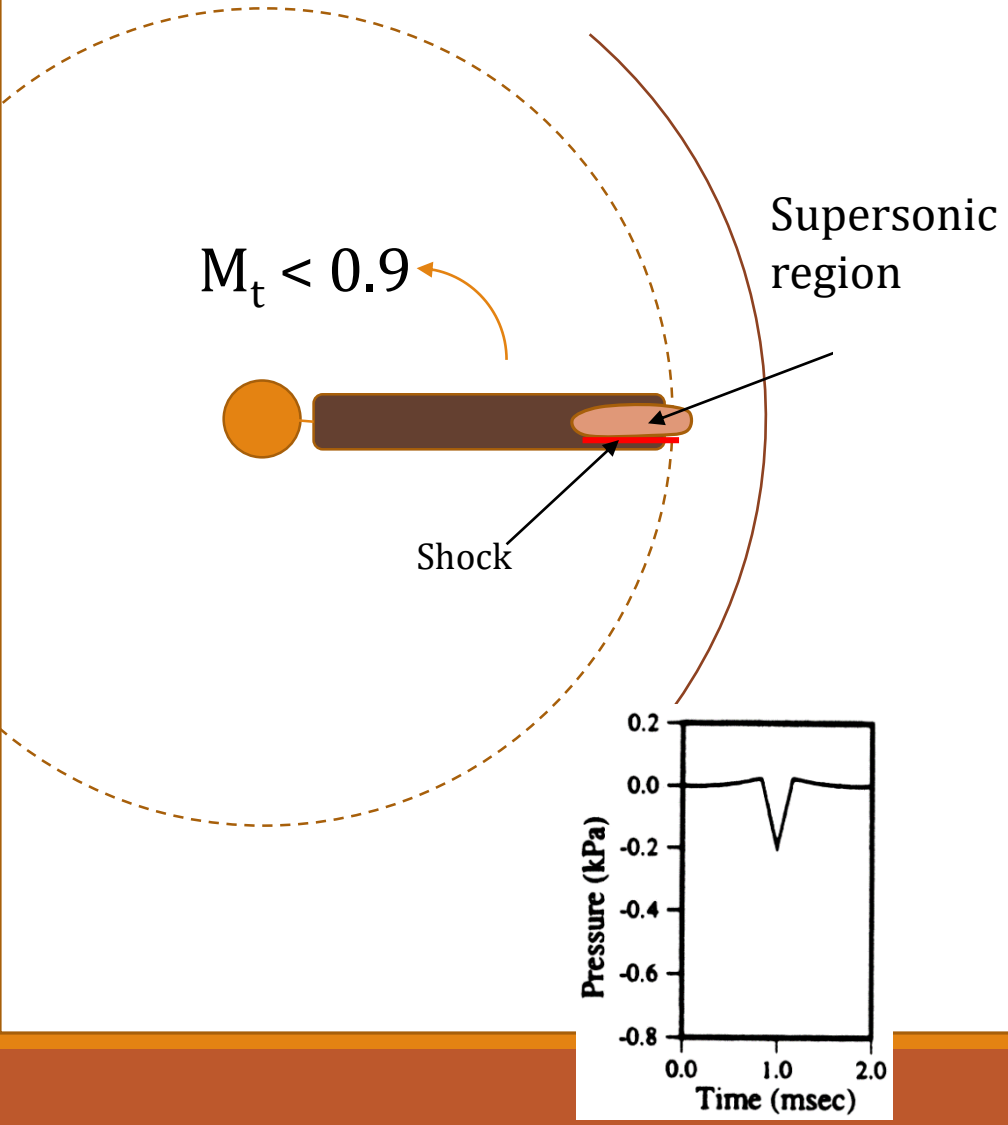


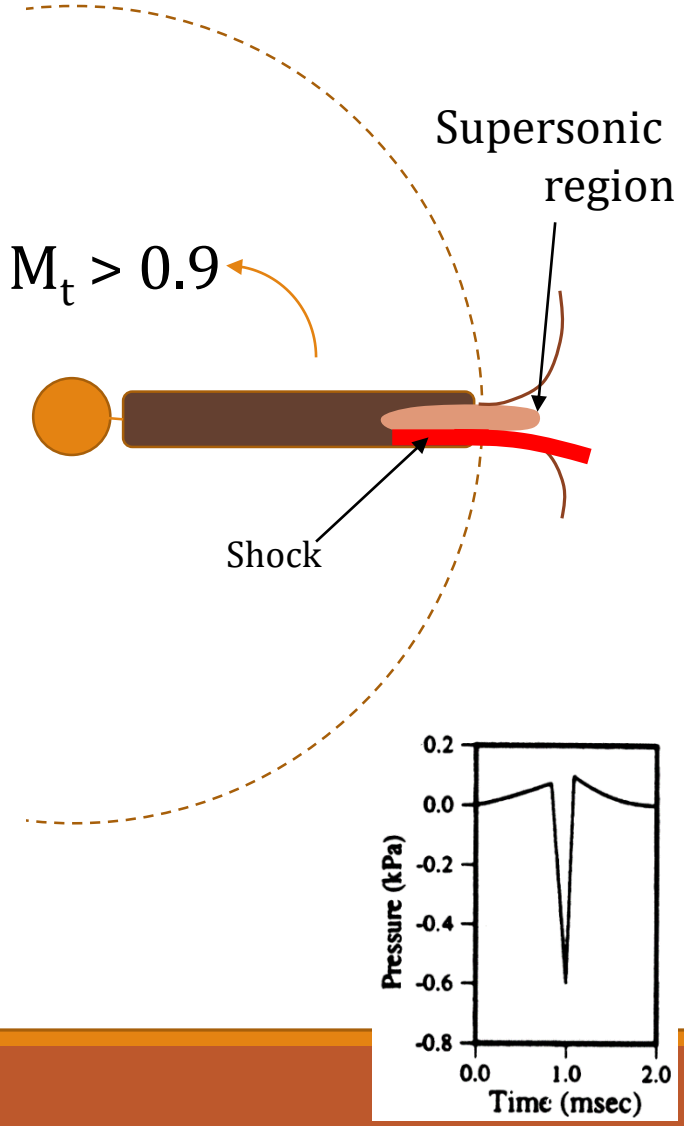
Trailing edge tabs can reduce pitching moments

Static and Dynamic Drag



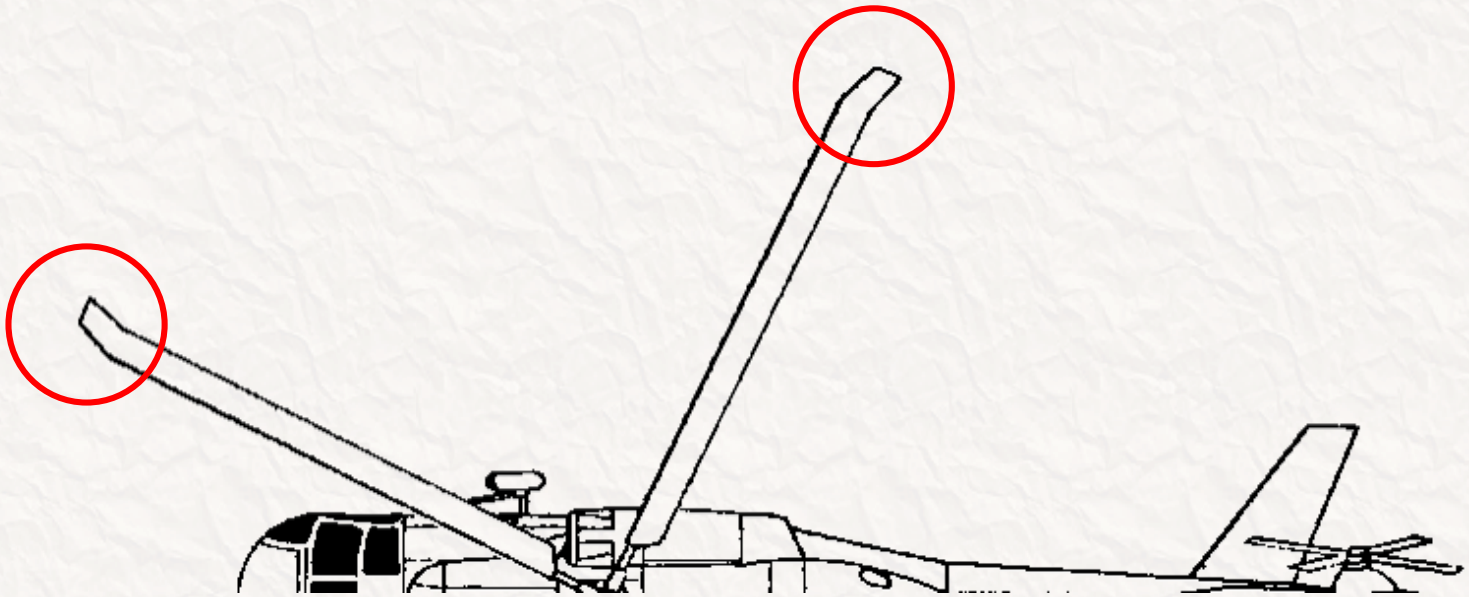
Shock near Blade Tip





Wave drag mitigation

- Thinner airfoils near tip
- Lower twist
- Backward sweep



UH- 60A (Black Hawk)

Design Perspective

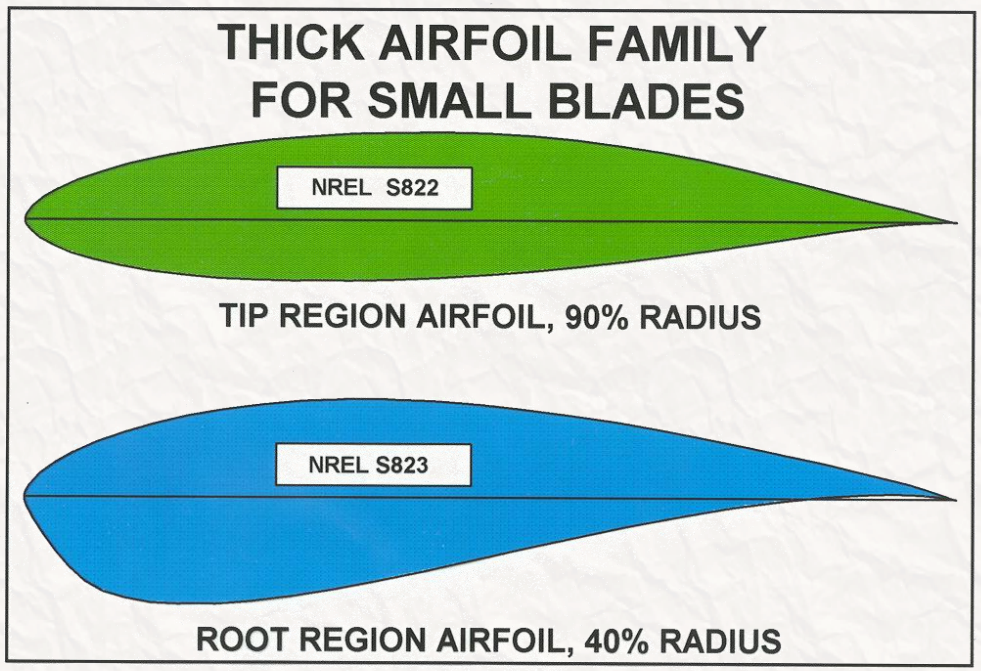
- A good Helicopter rotor airfoil
  ≠ good Wind Turbine rotor airfoil

- Operates in a very different environment
  - Airborne particles get stuck to the blades causing roughness effects (Difficult to clean blades regularly)
  - Huge structural and aerodynamic loading (in a 1.5 MW turbine, each blade weigh ~ 5 tons)

Design Philosophy and Criteria

- Specially-tailored airfoils for wind turbines
- Large thickness airfoils at the root. Thickness reduced closer to the tip
- Criteria
  - Moderate to high thickness ratio t/c
    - Rigid rotor: 16%–26% t/c
    - Flexible rotor: 11%–21% t/c
    - Small wind turbines: 10%-16% t/c
  - High lift-to-drag ratio
  - Minimal roughness sensitivity
  - Weak laminar separation bubbles

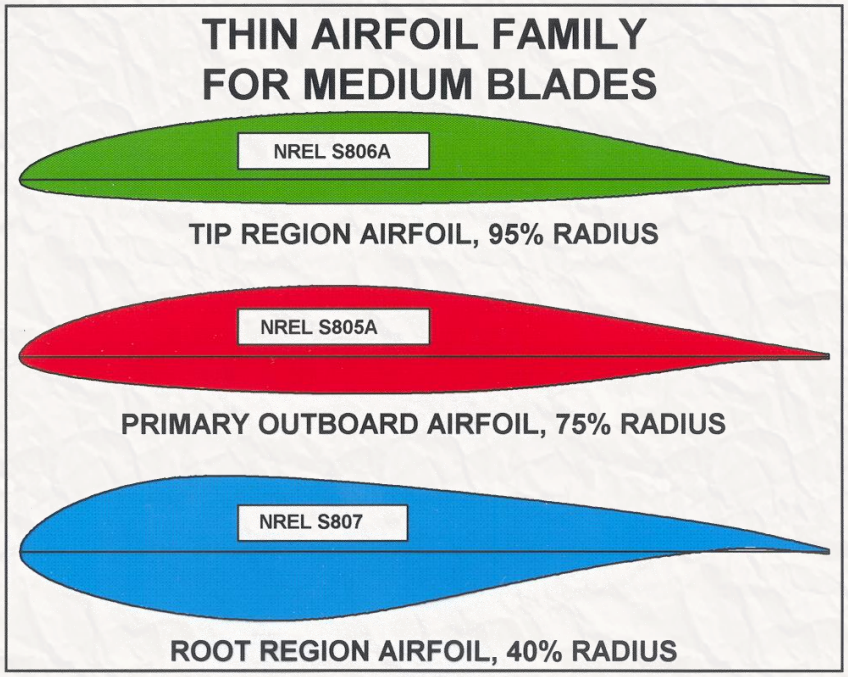
Airfoils Based on Rotor Sizes



Design Parameters

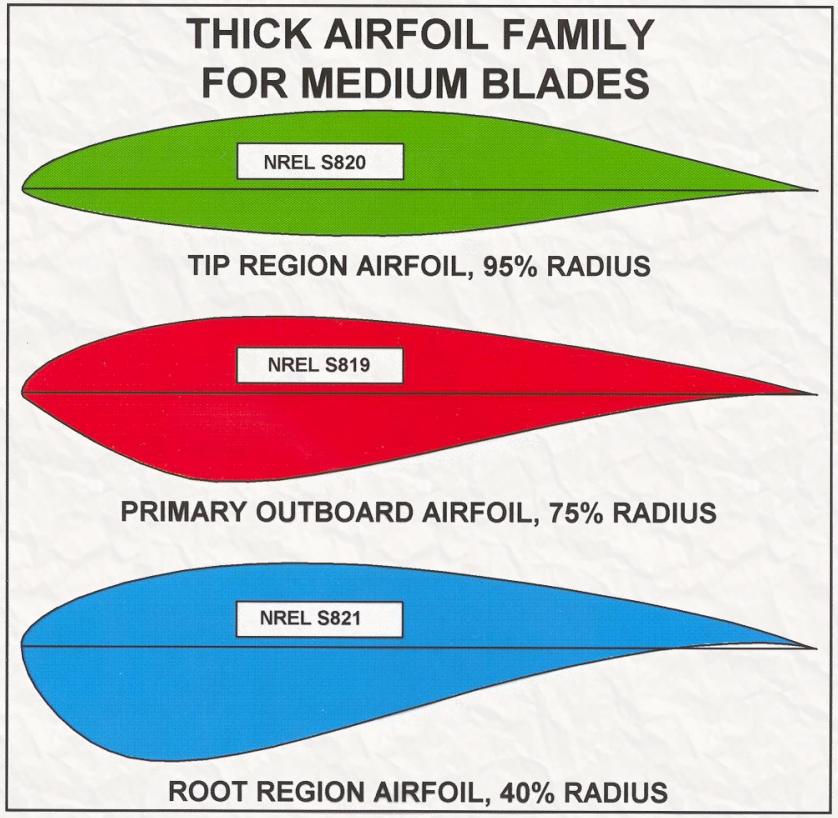
Airfoil	r/R	Re. No. ($x10^6$)	t/c	C_{lmax}	C_{dmin}	C_{mo}
S822	0.90	0.6	0.160	1.00	0.010	-0.07
S823	0.40	0.4	0.240	1.30	0.018	-0.15

Airfoils Based on Rotor Sizes



Design Parameters

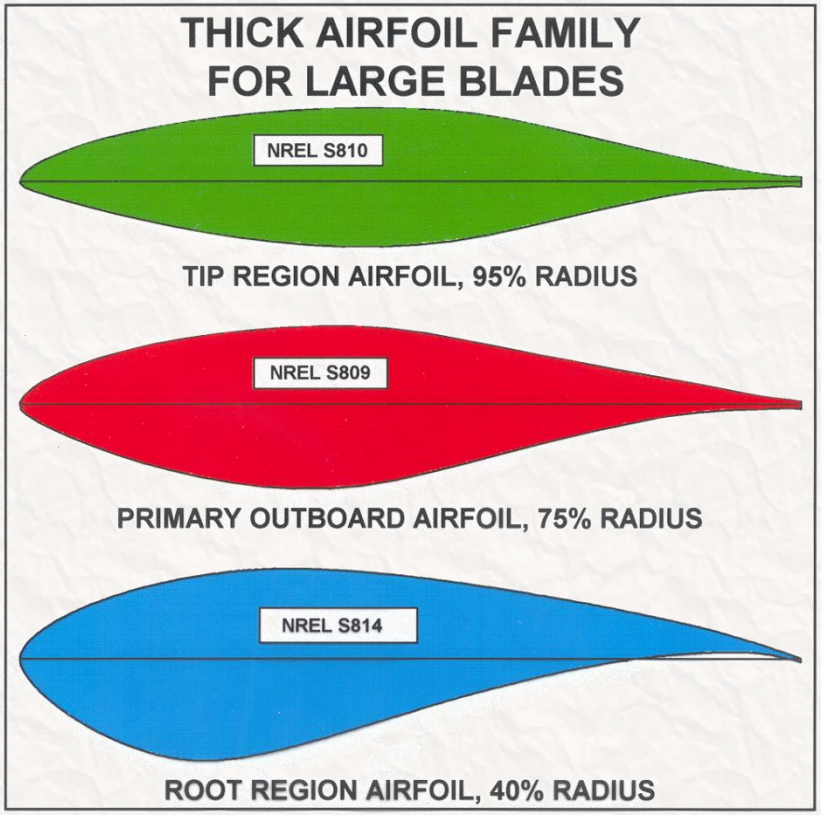
Airfoil	r/R	Re. No. ($x10^6$)	t/c	C_{lmax}	C_{dmin}	C_{mo}
S806A	0.95	1.3	0.115	1.10	0.004	-0.05
S805A	0.75	1.0	0.135	1.20	0.005	-0.05
S807	0.40	0.8	0.180	1.46	0.010	-0.10
S808	0.30	0.8	0.210	1.30	0.012	-0.12



Design Parameters

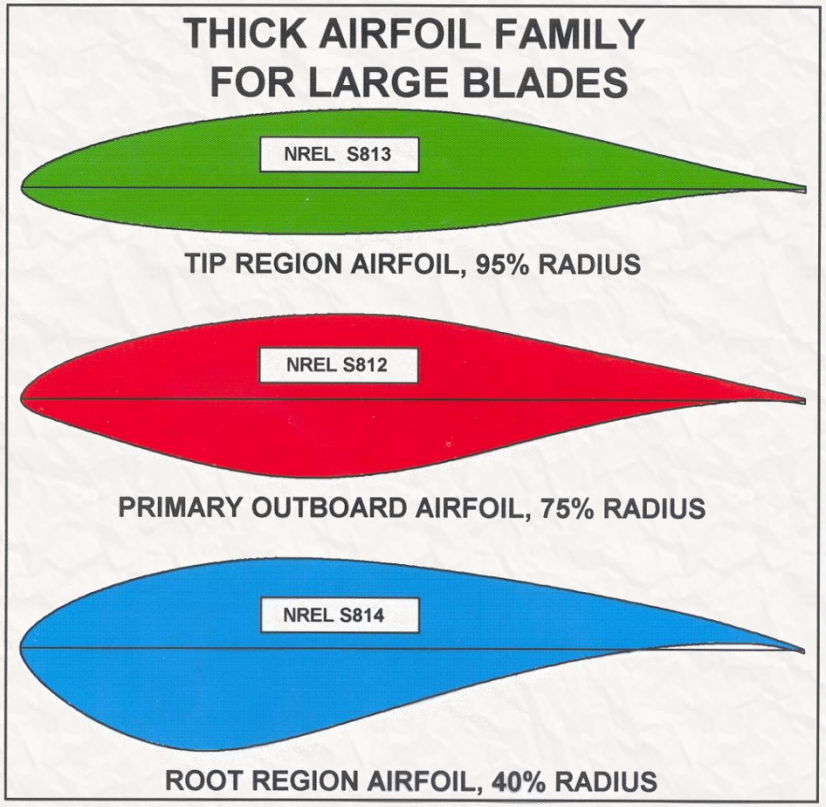
Airfoil	r/R	Re. No. ($x10^6$)	t/c	C_{lmax}	C_{dmin}	C_{mo}
S820	0.95	1.3	0.160	1.10	0.007	-0.07
S819	0.75	1.0	0.210	1.20	0.008	-0.07
S821	0.40	0.8	0.240	1.40	0.014	-0.15

Airfoils Based on Rotor Sizes



Design Parameters

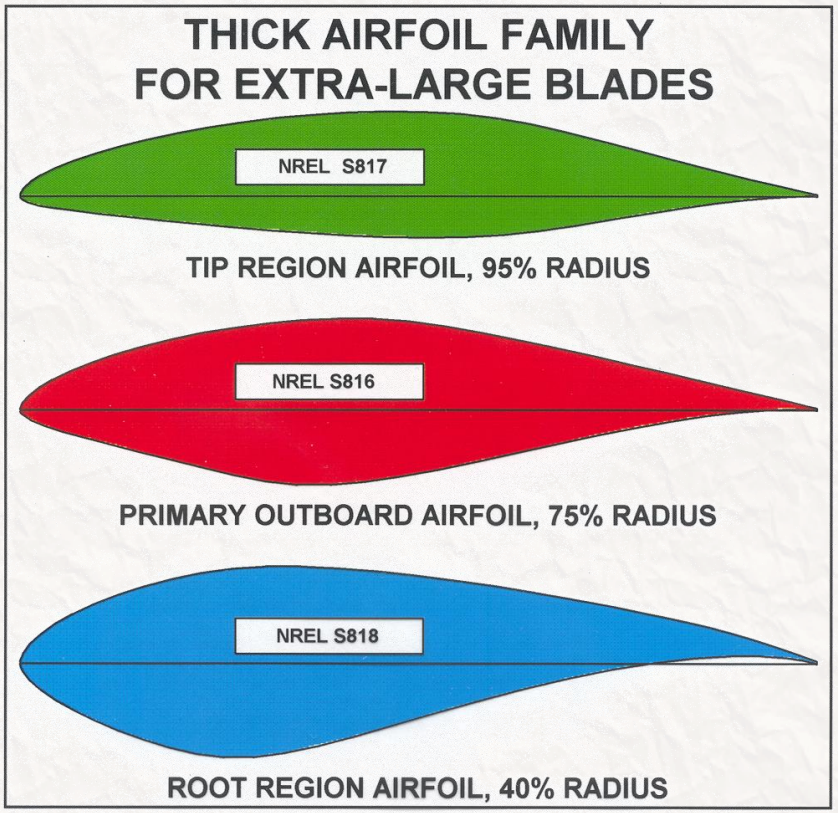
Airfoil	r/R	Re. No. ($x10^6$)	t/c	$C_{l_{max}}$	$C_{d_{min}}$	C_{m_o}
S810	0.95	2.0	0.180	0.90	0.006	-0.05
S809	0.75	2.0	0.210	1.00	0.007	-0.05
S814	0.40	1.5	0.240	1.30	0.012	-0.15
S815	0.30	1.2	0.260	1.10	0.014	-0.15



Design Parameters

Airfoil	r/R	Re. No. ($x10^6$)	t/c	$C_{l_{max}}$	$C_{d_{min}}$	C_{m_o}
S813	0.95	2.0	0.160	1.10	0.007	-0.07
S812	0.75	2.0	0.210	1.20	0.008	-0.07
S814	0.40	1.5	0.240	1.30	0.012	-0.15
S815	0.30	1.2	0.260	1.10	0.014	-0.15

Airfoils Based on Rotor Sizes



Design Parameters

Airfoil	r/R	Re. No. ($x10^6$)	t/c	$C_{l_{max}}$	$C_{d_{min}}$	C_{m_o}
S817	0.95	3.0	0.160	1.10	0.007	-0.07
S816	0.75	4.0	0.210	1.20	0.008	-0.07
S818	0.40	2.5	0.240	1.30	0.012	-0.15

Airfoil Selection

- Appropriate design Reynolds number
- Airfoil thickness: Centrifugal stiffening and desired blade rigidity
- Roughness Insensitivity